

# Genotypic and phenotypic profiles of antibiotic-resistant bacteria isolated from hospitalised patients in Bangladesh

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## Abstract

**OBJECTIVES** Characterisation of resistance phenotype and genotype is crucial to understanding the burden and transmission of antimicrobial resistance (AMR). This study aims to determine the spectrum of AMR and associated genes encoding aminoglycoside, macrolide and  $\beta$ -lactam classes of antimicrobials in bacteria isolated from hospitalised patients in Bangladesh.

**METHODS** 430 bacterial isolates from patients with respiratory, intestinal, wound infections and typhoid fever, presenting to clinical care from 2015 to 2019, were examined. They included *Escherichia coli* ( $n = 85$ ); *Staphylococcus aureus* ( $n = 84$ ); *Salmonella typhi* ( $n = 82$ ); *Klebsiella pneumoniae* ( $n = 42$ ); *Streptococcus pneumoniae* ( $n = 36$ ); coagulase-negative staphylococci ( $n = 28$ ); *Enterococcus faecalis* ( $n = 27$ ); *Pseudomonas aeruginosa* ( $n = 26$ ); and *Acinetobacter baumannii* ( $n = 20$ ). Reconfirmation of these clinical isolates and antimicrobial susceptibility tests was performed. PCR amplification using resistance gene-specific primers was done, and the amplified products were confirmed by Sanger sequencing.

**RESULTS** 53% of isolates were multidrug-resistant (MDR), including 97% of *Escherichia coli*. There was a year-wise gradual increase in MDR isolates from 2015 to 2018, and there was an almost twofold increase in the number of MDR strains isolated in 2019 ( $P = 0.00058$ ). Among the 5 extended-spectrum  $\beta$ -lactamases investigated, CTX-M-1 was the most prevalent (63%) followed by NDM-1 (22%); *Escherichia coli* was the major reservoir of these genes. The *ermB* (55%) and *aac(6')-Ib* (35%) genes were the most frequently detected macrolide and aminoglycoside resistance genes, respectively.

**CONCLUSION** MDR pathogens are highly prevalent in hospital settings of Bangladesh.

**keywords** acute respiratory infections, aminoglycoside, antimicrobial resistance, diarrhoea, extended-spectrum  $\beta$ -lactamase, macrolide, multidrug resistance, typhoid fever, wound infections

Sustainable Development Goals: Good Health and Wellbeing, Partnerships for the Goals

## Introduction

Antimicrobial resistance (AMR) is an emerging global problem. Nosocomial infections have been recognised to

have a high rate of multidrug-resistant (MDR) pathogens. It is estimated that 8.7% of hospitalised patients worldwide develop nosocomial infections, which are the leading cause of surgical failure, treatment failure,

increased costs and even deaths [1]. Numerous reports suggest that absolute numbers of infections due to resistant microbes are increasing globally [2–4], and may cause up to 10 million deaths/year – more than any other causes – by 2050. This problem is inequitably distributed, with approximately 90% of the predicted deaths estimated to occur in Asia and Africa [5]. Treatment of patients infected with resistant pathogens is associated with higher costs, requiring additional investigations and longer hospitalisation [6]. The overall crude economic burden of AMR was estimated to be at least €1.5 billion in 2007 in Europe and \$55 billion in 2000 in the United States, including patients and hospital costs [7]. Without appropriate action, by 2050 the global economy may lose more than USD 6 trillion annually because of AMR, which is nearly 4% of Gross Domestic Product (GDP) [6].

Studies show that low-income countries like Bangladesh are more affected by AMR because of the widespread misuse of antimicrobials, non-human antimicrobial use, poor quality of drugs, inadequate surveillance and factors associated with individual and national poverty indicators such as poor healthcare standards, malnutrition, chronic and repeated infections, and unaffordability of more effective and costly drugs [8, 9]. Research regarding AMR epidemiology may ultimately lead to interventions for AMR.

Regional surveillance programmes have tracked AMR in several countries, but many gaps still exist in middle- and low-income countries that lack the infrastructure for comprehensive monitoring [10]. A well-functioning national AMR surveillance system is vital to planning and implementing the national AMR strategy. Studies on antimicrobial resistance encompassing different aspects on a national scale are critical because they provide information on the extent of resistance rates, and detect emerging patterns of resistance. Understanding how resistance is changing over time is important for several reasons: (1) establishing prescribing guidelines, (2) determining investment in new therapies and (3) improving the targeting of campaigns to reduce antimicrobial resistance. It also provides a baseline for future analysis and comparison with other countries [11].

Previous studies on AMR in Bangladesh have largely focused on Gram-negative bacteria [12–14], and a small number of antimicrobial resistance genes [15–17]. Here, we investigated the contemporary antimicrobial resistance patterns in Bangladesh, including that of Gram-positive bacteria, which have emerged as global threats in serious infections such as pneumonia, wound infections and diabetic ulcers [18–20]. We investigated resistance genes that

had not been investigated in previous studies in Bangladesh, and examined trends of resistance over time.

## Materials and Methods

### Bacterial isolates

We analysed a total of 430 preserved bacterial isolates. These isolates were collected from hospitalised patients with acute respiratory infections (nasopharyngeal swab), wound infections (wound swab), typhoid fever (blood samples) and diarrhoeal diseases (stool samples) from four tertiary hospitals of Dhaka city namely Bangladesh University of Health Sciences, Dhaka Medical College, Uttara Adhunik Medical College Hospital, and Shaheed Suhrawardy Medical College and Hospital between 2015 and 2019. The strains were preserved at  $-70^{\circ}\text{C}$  as glycerol stock. Bacterial identification and confirmation were performed by routine conventional microbial cultures and biochemical tests using standard techniques [21].

### Antimicrobial susceptibility testing

Profiling of antimicrobial susceptibility/resistance was performed by the modified Kirby–Bauer disc diffusion method and the bacterial strains were identified as either sensitive or resistant to an antimicrobial based on the diameter of the zone of inhibition, according to Clinical and Laboratory Standard Institute (CLSI) guidelines. MDR strains were identified as resistant to three or more antimicrobial classes [22].

Cartridges of antimicrobial discs for Nalidixic acid (30  $\mu\text{g}$ ), Cotrimoxazole (25  $\mu\text{g}$ ), Ciprofloxacin (5  $\mu\text{g}$ ), Azithromycin (15  $\mu\text{g}$ ), Ampicillin (10  $\mu\text{g}$ ), Cefixime (5  $\mu\text{g}$ ), Gentamycin (10  $\mu\text{g}$ ), Chloramphenicol (30  $\mu\text{g}$ ), Ceftriaxone (30  $\mu\text{g}$ ), Imipenem (10  $\mu\text{g}$ ), Piperacillin–tazobactam (10  $\mu\text{g}$ ), Tetracycline (30  $\mu\text{g}$ ), Trimethoprim–sulfamethoxazole (25  $\mu\text{g}$ ), Rifampicin (5  $\mu\text{g}$ ), Cefoxitin (30  $\mu\text{g}$ ), Penicillin (10  $\mu\text{g}$ ), Vancomycin (30  $\mu\text{g}$ ), Linezolid (30  $\mu\text{g}$ ), Colistin (10  $\mu\text{g}$ ), Erythromycin (15  $\mu\text{g}$ ), Televancin (30  $\mu\text{g}$ ) and Clindamycin (2  $\mu\text{g}$ ) were obtained from Oxoid (Hampshire, UK). The cartridges were stored between 4 and  $20^{\circ}\text{C}$  and allowed to come to room temperature prior to use. After inoculation with the isolates and placement of the discs, Mueller–Hinton agar plates were incubated at  $37^{\circ}\text{C}$  for 24 h and the zones of inhibition were measured.

### PCR amplifications and sequencing

Genomic DNA of the isolated bacterial colonies was extracted using QIAamp DNA Mini Kit (Qiagen, Hilden,

Germany) following manufacturer's guidelines. DNA extracts that were positive for the 16S rRNA gene in PCR assays were examined for the presence of resistance genes for different classes of antimicrobials, namely aminoglycoside, macrolide and  $\beta$ -lactam. The gene sequences were retrieved from the nucleotide database of National Centre for Biotechnology Information (NCBI). The retrieved FASTA sequences were then used to design gene-specific primers using NCBI Primer-BLAST. The description of the primer sets used in this study for each resistance gene is depicted in Table S1. Gene-specific PCR amplification using each set of primers was carried out on a T100™ thermal cycler (Bio-Rad, California, United States). The PCR reaction volume was 10  $\mu$ L containing 1  $\mu$ L 10X PCR buffer, 0.3  $\mu$ L 50 mM  $MgCl_2$ , 0.2  $\mu$ L of 10 mM dNTPs mixture, 0.5  $\mu$ L forward and reverse primers, 0.05  $\mu$ L of Taq polymerase, 5.45  $\mu$ L Nuclease-free Water and 2  $\mu$ L of template DNA. The following cycling conditions were used: initial denaturation at 95 °C for 5 min, 35 cycles of denaturation at 95 °C for 30 s, annealing at 55 °C, cyclic extension at 72 °C for 45 s and final extension at 72 °C for 6 min. The amplified PCR products were visualised on a 1% agarose gel stained with SYBR Safe (Invitrogen, CA, USA). Later, the PCR products were purified using Quick PCR product purification kit (Invitrogen) following the manufacturer's guidelines.

### Sanger DNA sequencing

The PCR-amplified DNA was sequenced using ABI PRISM-310 software version 3.1.0 (Applied Biosystems, CA, USA). To prepare cycle sequencing reaction mixtures, 1 ng of purified DNA was added to a reaction solution containing 0.50  $\mu$ L Big dye terminator sequencing mix and 2.0  $\mu$ L of 5X PCR sequencing buffer with 0.2  $\mu$ L primer-solution. Then, the PCR strip was placed in the Mastercycler® gradient Thermal Cycler and subjected to the following thermal cycling profile: pre-denaturation at 95 °C for 10 min; 25 cycles of denaturation at 95 °C for 10 s, annealing at 55 °C for 5 s, cyclic extension at 72 °C for 4 min; and a final extension at 72 °C for 6 min.

### Data analysis

Sequencing data were analysed by Chromas Lite 2.4 software to identify the target sequence by alignment with the reference sequence. The obtained sequence was subjected to further analysis using Basic Local Alignment Search Tool (BLAST) for finding sequence similarity with sequences already reported in online databases. Graphs were generated

using GraphPad Prism v7 software (San Diego, CA, USA). Statistical analysis was performed using the XLSTAT version 2019.1.2. Proportions of MDR isolates and antimicrobial resistance trend over time were compared using the Cochran–Armitage test for trends, where the threshold for statistical significance was  $P < 0.05$ .

### Results

We analysed 430 clinical bacterial isolates from patients with acute respiratory infections ( $n = 40$ ), wound infections ( $n = 239$ ), typhoid ( $n = 82$ ) and diarrhoea ( $n = 69$ ). These clinical specimens were collected from patients admitted to various tertiary hospitals of Dhaka city between 2015 and 2019 (Table S2). We detected 8 of 12 global priority pathogens (GPP) according to WHO: *Acinetobacter baumannii* ( $n = 20$ ), *Pseudomonas aeruginosa* ( $n = 26$ ), *Klebsiella pneumoniae* ( $n = 42$ ) and *Escherichia coli* ( $n = 85$ ), all Priority 1 (Critical) pathogens, as well as *Staphylococcus aureus* ( $n = 84$ ), *Salmonella typhi* ( $n = 82$ ) and *Enterococcus faecalis* ( $n = 27$ ), Priority 2 (High) pathogens, and *Streptococcus pneumoniae* ( $n = 36$ ) isolates, a Priority 3 (Medium) pathogen. Although coagulase-negative staphylococci (CNS) is not enlisted in the GPP list, we included CNS ( $n = 28$ ) as these organisms represent an increasingly important cause of hospital-acquired infections [23, 24].

### Antimicrobial resistance phenotype

We determined the antimicrobial resistance patterns of the isolated Gram-negative (Table 1A) and Gram-positive (Table 1B) bacterial isolates. Among the Gram-negatives, *P. aeruginosa* and *K. pneumoniae* isolates exhibited high resistance to almost all the antibiotics tested (Table 1A). We found that across our collection of *P. aeruginosa* isolates, we detected resistance against all eleven antimicrobials. Within this collection, we detected high prevalence of resistance against seven antimicrobials including gentamycin (87%), azithromycin (80%), ampicillin (70%), ciprofloxacin (59%), ceftriaxone (45%), imipenem (45%) and piperacillin tazobactam (41%). In addition, we found that 89% of isolates of *S. typhi*, which is the causative agent for typhoid fever, were resistant to nalidixic acid.

For Gram-positive bacteria, we found >50% of *S. aureus* isolates resistant against ceftioxin, gentamycin, ciprofloxacin and rifampicin (Table 1B). In addition, 84% *S. aureus* isolates were found to be MRSA (Methicillin-resistant *S. aureus*) demonstrated through resistance to ceftioxin. *S. aureus* isolates showed less resistance against vancomycin (8% resistance), trimethoprim–sulfamethoxazole (6%) and chloramphenicol (no resistance).

**Table 1** Rates of resistance of the isolated organisms to different antimicrobials. (A) For Gram-negative organisms, (B) for Gram-positive organisms. Kirby–Bauer disc diffusion method was applied to determine resistance rate of each isolate

| A. Percentage resistant to antimicrobial for Gram-negative organisms. |     |     |     |     |     |     |    |     |     |     |    |     |     |
|-----------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|----|-----|-----|-----|----|-----|-----|
| Microorganisms                                                        | AMP | TZP | COT | CRO | CFM | IMP | CN | AZM | TET | CIP | C  | NA  | CL  |
| <i>Escherichia coli</i> (n = 85)                                      | 89  | 21  | N/A | 40  | 43  | 56  | 51 | 58  | 20  | 36  | 7  | N/A | 0   |
| <i>Salmonella typhi</i> (n = 82)                                      | 18  | N/A | 9   | 0   | 0   | 0   | 32 | 11  | N/A | 10  | 9  | 89  | N/A |
| <i>Pseudomonas aeruginosa</i> (n = 26)                                | 70  | 41  | N/A | 45  | 37  | 45  | 87 | 80  | 34  | 59  | 23 | N/A | 11  |
| <i>Klebsiella pneumoniae</i> (n = 42)                                 | 97  | 17  | N/A | 50  | 66  | 14  | 72 | 77  | 7   | 67  | 21 | N/A | 10  |
| <i>Acinetobacter baumannii</i> (n = 20)                               | 19  | 0   | N/A | 34  | 17  | 0   | 42 | 0   | 0   | 9   | 27 | N/A | 0   |

| B. Percentage resistant to antimicrobial for Gram-positive organisms. |     |     |     |      |     |      |      |    |    |     |    |     |     |    |
|-----------------------------------------------------------------------|-----|-----|-----|------|-----|------|------|----|----|-----|----|-----|-----|----|
| Microorganisms                                                        | VAN | FOX | TEL | CN   | AZM | TET  | CIP  | CD | C  | RIF | LZ | SXT | PEN | E  |
| <i>Staphylococcus aureus</i> (n = 84)                                 | 8   | 84  | 19  | 54   | 46  | 32   | 53.5 | 50 | 0  | 61  | 24 | 6   | 66  | 54 |
| CNS (n = 28)                                                          | 6   | 41  | 25  | 18   | 55  | 31   | 68   | 38 | 10 | 38  | 35 | 12  | 66  | 58 |
| <i>Enterococcus faecalis</i> (n = 27)                                 | 19  | 0   | 16  | 12.5 | 0   | 12.5 | 38   | 12 | 8  | 19  | 19 | 12  | 56  | 60 |
| <i>Streptococcus pneumoniae</i> (n = 36)                              | 9   | N/A | 12  | 25   | 46  | 11   | 3    | 25 | 0  | 10  | 3  | 0   | 36  | 50 |

AMP, ampicillin; AZM, azithromycin; AZM, azithromycin; C, chloramphenicol; C, chloramphenicol; CD, clindamycin; CFM, cefixime; CIP, ciprofloxacin; CIP, ciprofloxacin; CL, colistin. VAN, vancomycin; CN, gentamycin; CN, gentamycin; COT, cotrimoxazole; CRO, Ceftriaxone; E, erythromycin; FOX, cefoxitin; IMP, imipenem; LZ, linezolid; NA, nalidixic acid; PEN, penicillin; RIF, rifampicin; SXT, trimethoprim–sulfamethoxazole; TEL, telavancin; TET, tetracycline; TET, tetracycline; TZP, Piperacillin–tazobactam.

Colistin resistance was detected in 10% of *K. pneumoniae* and 11% of *P. aeruginosa* isolates.

To evaluate the extent of multidrug resistance, the percentage of resistance for each bacterial species was assessed. Approximately 53% of isolates were identified as MDR. Figure 1a shows that *E. coli* (97%) was found to be the most prevalent MDR organism, followed by *S. aureus* (90%), *P. aeruginosa* (83%), *K. pneumoniae* (81%), CNS (76%), *S. aureus* (75%), *A. baumannii* (67%), *Enterococcus faecalis* (60%), *S. pneumoniae* (56%) and *S. typhi* (35%). Next, in Figure 1b, we found that resistance to more than 6 antimicrobials was 68% for *S. aureus*, followed by 62% for *P. aeruginosa*. Among *P. aeruginosa* and *K. pneumoniae*, only 2% of isolates were sensitive to all antimicrobial tested.

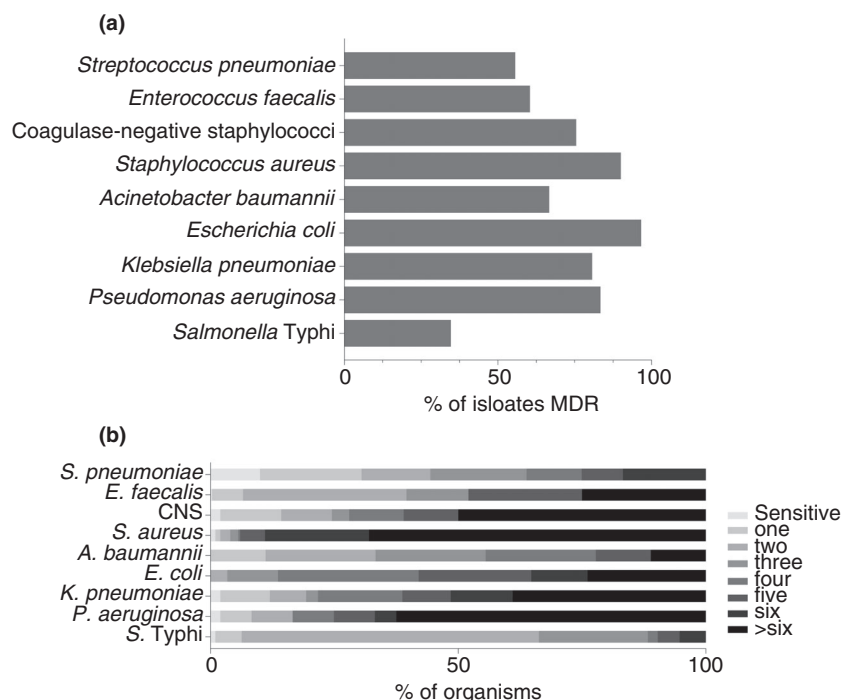
### Occurrence of antimicrobial resistance genes

We next determined the possible resistance genes harboured by these organisms by sequencing the PCR amplicons towards a variety of known resistance genes. We found that *E. coli*, *P. aeruginosa* and *K. pneumoniae* strains harboured almost all of the resistance genes analysed (Figure 2). We detected that *S. aureus* and *P. aeruginosa* isolates also had high percentage of resistance genes, in particular *ermB*, *aac(6')-Ib* and *aac(3)-II*. The most frequently identified ESBL gene was *CTX-M-1* (Amber class-A type) with the highest frequencies found in *E. coli*

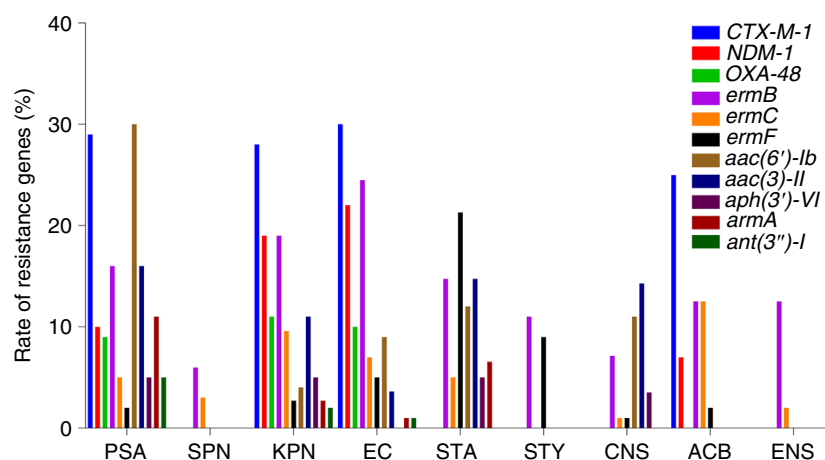
(30%), followed by *P. aeruginosa* (29%) and *K. pneumoniae* (28%). *NDM-1*, which is a metallo- $\beta$ -lactamase of Amber class-B, was also commonly detected. The predominant bacterial species which harboured this gene was *E. coli* (22%). The commonly detected macrolide resistance gene, *ermB* was detected in 24% *E. coli*, 19% *K. pneumoniae* and 16% *P. aeruginosa*. 30% of *P. aeruginosa* and 12% *S. aureus* were found to possess the *aac(6')-Ib* gene, which is a frequently detected aminoglycoside resistance gene.

### Increase in AMR gene prevalence over time

To examine the resistance trend over time of the pathogenic organisms listed in the WHO priority list, we compared the year-wise percentage of MDR organisms along with the resistance genes between 2015 and 2019. We found increases in percentage of organisms harbouring *CTX-M-1*, *ermB*, *aac(6')-Ib*, *NDM-1* and *aac(3)-II* resistance genes over time (Figure 3a). A number of resistance genes that were not detected in 2015, including *OXA-48*, *ermC*, *ermF*, *aph(3')-VI* and *armA*, appeared during the study period (Figure 3a). Significant differences (all  $P < 0.005$ ) were found in case of ESBL, macrolide, aminoglycoside resistance Gene-positive strains over the 5-year period (Table 2). A similar significant trend had also been observed for bacterial strains harbouring *NDM-1*, *CTX-M-1*, *aac(6')-Ib*, *aac(3)-II* and *ermB* genes



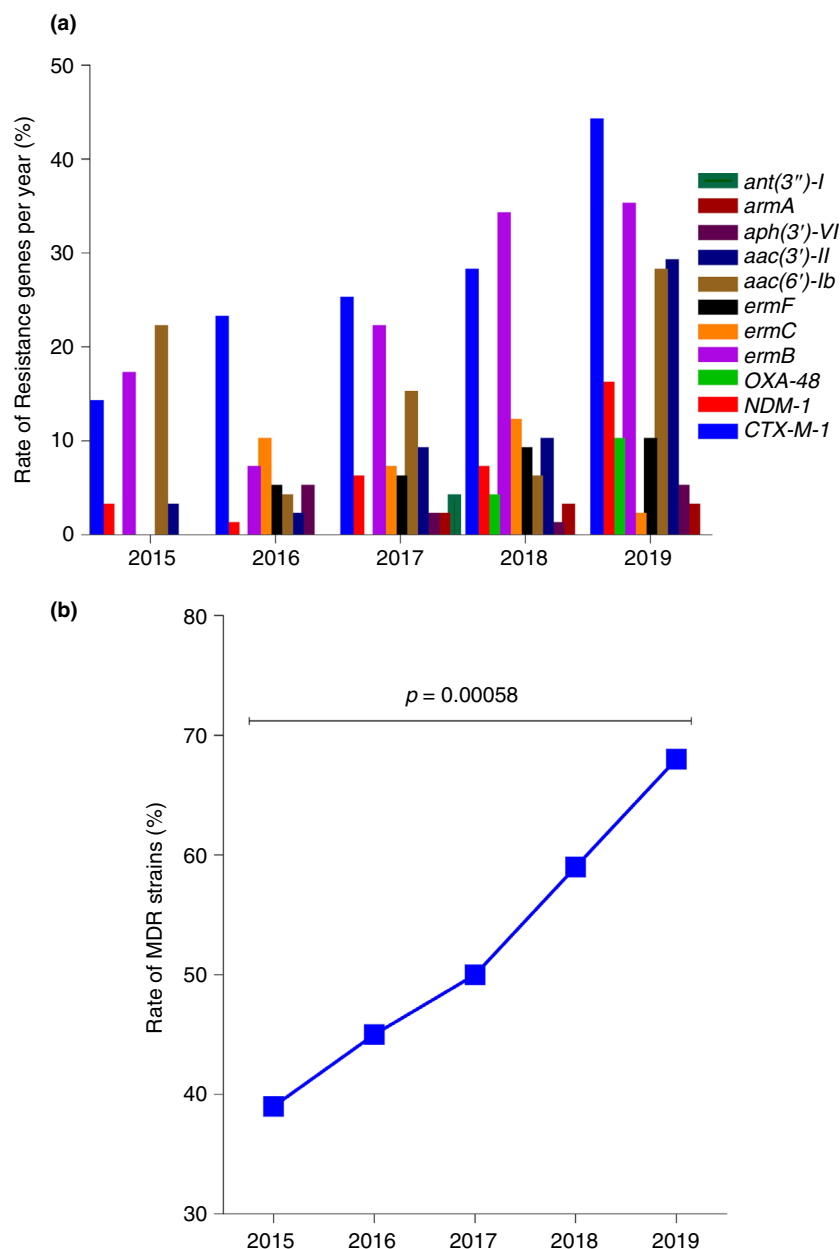
**Figure 1** Resistance frequency of the indicated clinical isolates. (a) MDR frequencies of the indicated organisms. The percentage of isolates displaying resistance to more than three antimicrobial classes (b) Proportion of isolates resistant to the indicated number of drugs. Each shade of colour indicates the number of antimicrobials to which a bacterial species was resistant.



**Figure 2** Distribution of antimicrobial resistance genes in pathogens. In the Figure, PSA, *Pseudomonas aeruginosa* ( $n = 26$ ); SPN, *Streptococcus pneumoniae* ( $n = 36$ ); KPN, *Klebsiella pneumoniae* ( $n = 42$ ); EC, *Escherichia coli* ( $n = 85$ ); STA, *Staphylococcus aureus* ( $n = 84$ ); STY, *Salmonella Typhi* ( $n = 82$ ); ACB, *Acinetobacter baumannii* ( $n = 20$ ); ENS, *Enterococcus faecalis* ( $n = 27$ ) and CNS, coagulase-negative staphylococci ( $n = 28$ ).

over the years. Our comparisons only applied to those 5 genes that were consistently detected throughout the 5 years. The proportion of MDR organisms also increased over time (Figure 3b): from 39% (31/80) in

2015 increasing to 68% (62/91) in 2019 ( $P = 0.00058$ ). Overall, the data reveal that rates of MDR organisms and presence of resistance genes have significantly increased over time.



**Figure 3** Antimicrobial resistance genes detected in bacterial isolates over time. (a) Comparison of the percentage of the organisms carrying antimicrobial resistance genes. (b) Comparison of the rate of MDR organisms (resistant to more than three antimicrobial classes) that included PSA, SPN, KPN, EC, STA, STY, CNS, ACB and ENS isolates between 2015 and 2019. For a and b, the number of samples analysed was 80, 84, 85, 90 and 91 for 2015, 2016, 2017, 2018 and 2019, respectively. Each colour in a indicates an antimicrobial resistance gene. Cochran–Armitage trend test was performed for analysis of the trend of MDR strains.  $P < 0.05$  was considered significant.

#### Coexistence of multiple antimicrobial resistance genes in the tested isolates

Next, we wanted to investigate co-presence of resistance genes. Out of 430 clinical strains, a total of 185 isolates

revealed to possess multiple resistance genes and there were 26 different kinds of coexistence of resistance genes (Table S3). *E. coli* ( $n = 67$ ), *K. pneumoniae* ( $n = 37$ ) and *S. aureus* ( $n = 34$ ) strains were found to be the

**Table 2** Percentage of NDM-1, CTX-M-1, *ermB*, *aac(6')-Ib*, *aac(3)-II*, ESBL, macrolide and aminoglycoside resistance Gene-positive strains per year. *P* values are calculated with Cochran–Armitage trend test

| Antimicrobial resistance genes | Year (Number of strains) |           |           |           |           | Cochran's <i>Q</i> value | <i>P</i> value |
|--------------------------------|--------------------------|-----------|-----------|-----------|-----------|--------------------------|----------------|
|                                | 2015 (80)                | 2016 (84) | 2017 (85) | 2018 (90) | 2019 (91) |                          |                |
| NDM-1                          | 4                        | 2         | 7         | 8         | 18        | 40.33                    | 0.0017         |
| CTX-M-1                        | 13                       | 22        | 24        | 27        | 43        | 71.37                    | 0.00025        |
| <i>ermB</i>                    | 16                       | 7         | 22        | 34        | 35        | 67.69                    | 0.000013       |
| <i>aac(6')-Ib</i>              | 22                       | 4         | 15        | 6         | 28        | 58.38                    | 0.0005         |
| <i>aac(3)-II</i>               | 3                        | 2         | 9         | 10        | 29        | 70.66                    | <0.0000001     |
| ESBL                           | 40                       | 32        | 26        | 38        | 55        | 71.111                   | 0.0019         |
| Macrolide                      | 29                       | 23        | 29        | 31        | 36        | 33.696                   | 0.00025        |
| Aminoglycoside                 | 34                       | 26        | 25        | 29        | 37        | 36.823                   | 0.001          |

predominant organisms in harbouring the co-resistance genes. In the 85 *E. coli* isolates analysed, coexistence of CTX-M-1 and *ermB* ( $n = 18$ ) was identified as the most prevalent combination. One *E. coli* isolate possessed a set of five genes, namely *ermB*, *ermF*, *aac(6')-Ib*, *ant(3'')-I* and CTX-M-1. All 26 (100%) *P. aeruginosa* isolates that had been analysed showed co-presence of resistance genes in different combinations. A *P. aeruginosa* isolate was seen to contain multiple ESBL genes (CTX-M-1, NDM-1 and OXA-48), with the addition of *aac(6')-Ib*, an aminoglycoside resistance gene. Overall, the data reveal that a large number of Gram-negative *E. coli*, *P. aeruginosa* and *K. pneumoniae* isolates had co-presence of resistance genes. On the other hand, only *S. aureus* isolates among the Gram-positive organisms had significant co-presence of resistance genes.

## Discussion

The present study characterises the antimicrobial resistance spectrum of the Gram-positive and Gram-negative bacterial isolates from patients in Bangladesh with wound infections, acute respiratory infections, diarrhoea and typhoid fever. Data on antimicrobial resistance, in particular genotypic characterisations, in Bangladesh have been scarce. Such knowledge can be used to inform public health evaluation and targeting of interventions to help control spread of antimicrobial-resistant bacteria.

The study found that 7 of the 12 pathogens from the WHO GPP list were isolated from wound infections, with high rates of ESBL producing Gram-negative organisms. This finding is in line with previous studies from Bangladesh [25]. A large number of wound patients, including patients with burns, are managed in tertiary care hospitals. In spite of technological advances that have been made in surgery and wound management, wound

infections are at high risk of nosocomial infections, especially in patients undergoing surgery [26].

The proportion of isolates which were MDR nearly doubled in the 5-year study period, consistent with findings from prior studies in Bangladesh [27], India [28] and Pakistan [29]. With exception of *S. pneumoniae*, *E. faecalis* and *A. baumannii*, most of the isolates exhibited alarming levels of resistance against the majority of the available first line of antimicrobials. Furthermore, *S. aureus*, *K. pneumoniae* and *P. aeruginosa* displayed resistance not only against all the first line of antimicrobials but also against most of the second-line antimicrobials tested. Our phenotypic findings are in conformity with the findings of multiple resistance genes in the organisms. Such highly resistant organisms may provide a conducive niche for horizontal transfer of plasmids carrying resistance genes [30], and such a niche may have potential for biofilm formation by the resident bacteria, which in turn may help horizontal gene transfer [31].

Our study revealed that colistin, the last resort antimicrobial, was the most active antimicrobial agent against the majority of Gram-negative bacteria, although colistin-resistant pathogens including *K. pneumoniae* and *P. aeruginosa* were detected. We also showed that isolates had relatively higher sensitivity to antimicrobials that were previous first-line agents, but more recently had decreased in use, such as chloramphenicol and trimethoprim–sulfamethoxazole, consistent with findings from a single-center study in India [11]. These changes are likely due to the replacement of these drugs as empiric treatment options with newer drugs such as macrolides and  $\beta$ -lactams. The findings suggest that older agents such as chloramphenicol and trimethoprim–sulfa may reemerge as options for empiric treatment, especially if resistance to macrolides, quinolones, and  $\beta$ -lactams continues to increase.

Our study revealed CTX-M-1 to be the most common ESBL resistance gene (63%), followed by NDM-1 (22%) and OXA-48 (15%). This is in agreement with a 2013-study conducted in Bangladesh which identified CTX-M-1 in 50% of the cases [32]. *E. coli* was found to be the main reservoir of CTX-M-1 gene. CTX-M-mediated genes have also been extensively detected in wild birds and poultry [33], and environmental water [12] in Bangladesh. This provides a vicious cycle of such resistance genes in environment, poultry and human. The finding strengthens the previous report demonstrating CTX-M-producing *Enterobacteriaceae* (CPE) as a major cause of both community-onset and hospital-acquired infections in Asian countries, including Korea [34], Taiwan [35], China [36], Hong Kong [37], Japan [38], Malaysia [39] and Thailand [40], and the incidence of serious infections due to CPE likely will continue to increase [41].  $\beta$ -lactam antimicrobials, especially cefotaxime, is frequently used as first-line therapy, which is leading to an alarming escalation of CTX-M-mediated resistance. Similarly, rates of NDM-1 increased between 2015 and 2019, similar to what has been demonstrated in India [42], Bangladesh [16] and Pakistan, even in community-onset infections [43]. This scenario of high NDM-1-producing organism in Bangladesh is of great concern because there are very few new antimicrobials in the pharmaceutical pipeline that are active against NDM-1 producers [44].

Also, among the macrolide resistance genes, *ermB* (55%) was the most prevalent, which is mostly associated with azithromycin resistance. In Bangladesh, many infectious diseases such as pneumonia, diarrhoea and typhoid are treated with azithromycin, an antimicrobial readily available over the counter. The observed frequent detection of *ermB* appeared concordant with results from studies reporting *ermB* detected from *S. Typhi* [14], *S. pneumoniae* [15] and *Shigella sonnei* [45]. Notably, we also detected other macrolide resistance genes such as *ermC* and *ermF*.

Co-presence of multiple resistance genes is a big concern in the spread of antimicrobial resistance. Our study found 43% strains harbouring more than one resistance genes, with an *E. coli* isolate possessing a set of five resistance genes. A study identified more than one carbapenemase gene in 45% of the clinical isolates which had been collected from a hospital in Dhaka [17]. Another study from our neighbouring country, Pakistan, found out that majority of NDM-1-producing *Enterobacteriaceae* co-produced CTX-M-1 and *armA*-type methylase [46]. These all prove that accumulation of multiple antimicrobial resistance genes in the clinical isolates raises an alarm on potential spread of extensively drug-resistant organisms in healthcare settings.

Finally, our data showed many antimicrobials with little efficacy against disease-specific pathogens that have acquired resistance genes, especially multiple resistance genes. Not only should these antimicrobials be discouraged to use for therapeutic purposes, there should be strict regulation on which antimicrobials should be available in the market and which should not be [47, 48]. It is hypothesised that resistant pathogens may lose resistance gene(s) without exposure to the antimicrobials for a relatively longer period of time and this might be the mechanism for regaining the efficacy by many antimicrobials that were in use previously but discontinued due to higher rates of non-susceptibility.

The study had a number of limitations. We examined only ESBL, macrolide and aminoglycoside resistance genes. Also, our isolates were de-identified, and thus, we were unable to associate our findings with clinical variables such as antimicrobial use history. Moreover, our surveillance was limited to hospitals in one city, and may not be generalisable to other areas, though our findings are consistent with prior studies from other parts of South Asia. Further investigations would benefit from whole genome sequencing to determine the complete resistance profile in these isolates. However, this remains a challenge in resource-limited settings.

## Conclusion

We detected increasing rates of MDR and resistance genes from bacteria isolated from patients in tertiary hospitals of the capital of Bangladesh. This has public health implications on the treatment and prevention of nosocomial infections. These observations reflect the overuse and misuse of antibiotics and underscore the need for urgent steps to stop the increasing incidence of resistance to the antibiotics in the hospital settings. Targeted efforts at decreasing inappropriate antibiotic use, as well as practices to prevent spread of nosocomial infections, are urgently needed.

## Acknowledgements

The authors thank Nirod Chandra Shaha, Statistician, International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b), and Polash Chandra Karmakar, Mst. Noorjahan Begum, Mohabbat Hossain, Fatema Ahmed Konica, Md Neamat Ullah, Subarna Sayed Ety, Syed Raiyanuzzaman, and Jeba Atkia Maisha for providing technical support. This study was funded by The World Academy of Sciences (TWAS), Research Grant number 17-555.



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### Supporting Information

Additional Supporting Information may be found in the online version of this article:

**Table S1** List of Extended-spectrum  $\beta$ -lactamase (ESBL), aminoglycoside and macrolide resistance gene specific primers selected for the study.

**Table S2** Information of clinical isolates analyzed.

**Table S3** Number of tested organisms harboring co-resistance genes.

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